Baseline Results

Reliable Hybrid SLC/MLC ReRAM Crossbars for Transformers

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1 What the baseline is

The baseline is **HyFlexPIM** (Song *et al.*, ISCA 2025), reproduced on our cell-level crossbar simulator. Every encoder linear layer of BERT-Base (72 matrices) and DistilBERT (36 matrices) is factorised with SVD, keeping $k = mn/(m+n)$ components so that storage is unchanged, and fine-tuned on SST-2 for 8 epochs. Components are ranked by the HyFlexPIM gradient score, $\sum_b |\partial L_b/\partial \sigma_r|$ over batches of 32 training sentences, and the top 20% of each layer go to SLC cells, the rest to MLC cells. Three reference chips are evaluated alongside it:

- **All SLC**: every component in reliable 1-bit cells (14 cells per weight), the most robust and most expensive chip;
- **All MLC**: every component in dense 2-bit cells (8 cells per weight), the cheapest chip;
- **Random 20%**: a random 20% of components in SLC, a control for whether the choice matters.

The hybrid chips use 9.2 cells per weight, 34% fewer than all-SLC. Accuracy is measured on all 872 SST-2 validation sentences and averaged over several simulated chips (random draws of all cell errors). Every placement is tested on the same chips. Errors are modelled per cell: 8-bit weights with one scale per output column, differential cell pairs, 7 SLC or 4 MLC bit slices per sign, and programming noise $\tilde{G} = G + \sigma G_{\max}\varepsilon$ on every cell.

2 Accuracy before hardware errors

Table 1: SST-2 accuracy (%) without hardware errors.

Model	FP32	INT8	SVD, truncated	SVD, fine-tuned
BERT-Base	92.43	91.86	89.33	90.14
DistilBERT	91.06	90.37	89.68	89.11

The SVD factorisation costs about two points after fine-tuning; this cost is common to every placement. On an error-free simulated chip every placement stays within about one point of the digital SVD model, confirming the simulator.

3 Programming noise

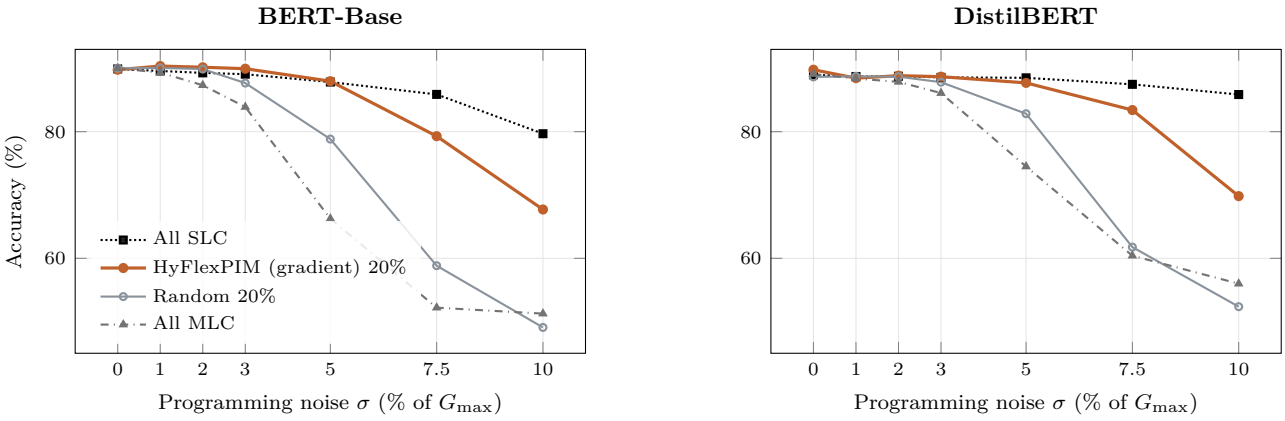


Figure 1: Accuracy against programming noise, mean of 5 chips.

Table 2: Accuracy (%) against programming noise σ, mean of 5 chips.

Model	Placement	0	1%	2%	3%	5%	7.5%	10%
BERT-Base	All SLC	89.9	89.6	89.3	89.1	87.8	85.9	79.7
	HyFlexPIM (gradient) 20%	89.8	90.4	90.2	90.0	88.0	79.3	67.7
	Random 20%	90.0	90.1	89.9	87.7	78.8	58.8	49.1
	All MLC	90.0	89.3	87.3	83.9	66.3	52.2	51.3
DistilBERT	All SLC	89.0	88.7	88.8	88.6	88.5	87.5	85.9
	HyFlexPIM (gradient) 20%	89.8	88.5	88.9	88.7	87.7	83.4	69.8
	Random 20%	88.6	88.7	88.7	87.8	82.8	61.8	52.4
	All MLC	89.1	88.4	87.9	86.1	74.5	60.4	56.0

- **All-MLC fails at moderate noise**: BERT-Base falls to 66.3% at $\sigma = 5\%$ and to near chance (52.2%) at 7.5%. A hybrid layout is necessary.
- **The HyFlexPIM hybrid works at low noise but degrades at high noise**: with 20% SLC it matches all-SLC up to $\sigma = 5\%$, then trails it by 6.6 points at 7.5% and 12 points at 10% on BERT-Base.
- **Random selection recovers little**: which components are protected matters, not only how many.

4 SLC share

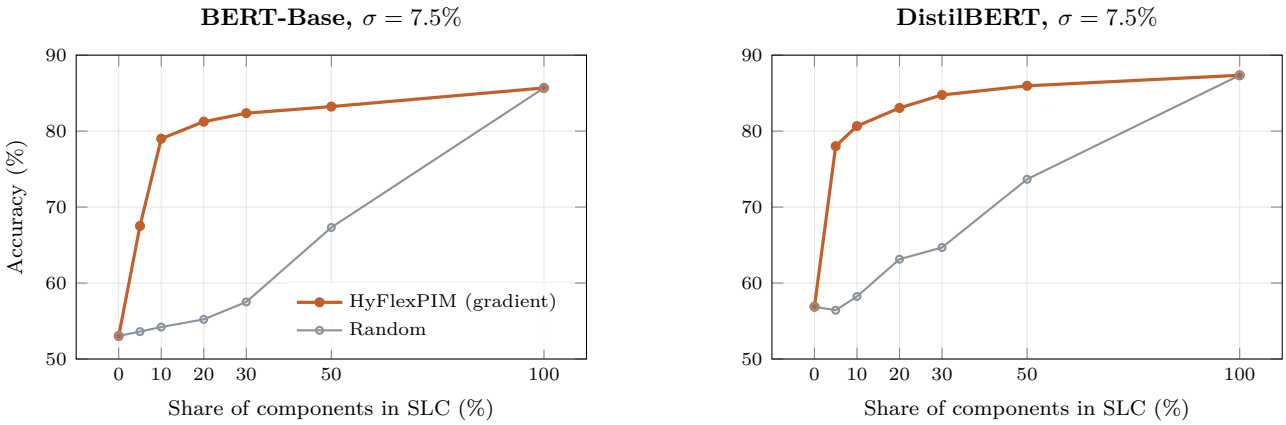


Figure 2: Accuracy against the share of components in SLC at $\sigma = 7.5\%$, mean of 20 chips; 0% is all-MLC and 100% all-SLC.

Table 3: Accuracy (%) of the HyFlexPIM gradient placement by SLC share, mean of 20 chips.

Model	σ	0% (all MLC)	5%	10%	20%	30%	50%	100% (all SLC)
BERT-Base	3%	82.1	88.7	90.0	90.0	89.9	89.6	89.2
	5%	66.8	84.2	88.2	88.4	88.4	88.1	88.0
	7.5%	53.0	67.5	79.0	81.2	82.4	83.2	85.7
DistilBERT	3%	86.0	88.4	88.5	88.6	88.8	88.7	88.7
	5%	74.8	87.3	87.3	87.7	88.1	87.8	88.2
	7.5%	56.9	78.0	80.7	83.1	84.8	86.0	87.4

At $\sigma = 7.5\%$ the gradient placement does not reach all-SLC even with 50% of the components in SLC on BERT-Base (83.2% against 85.7%), so most of the cell savings of the hybrid are lost at high noise.

5 Stuck cells

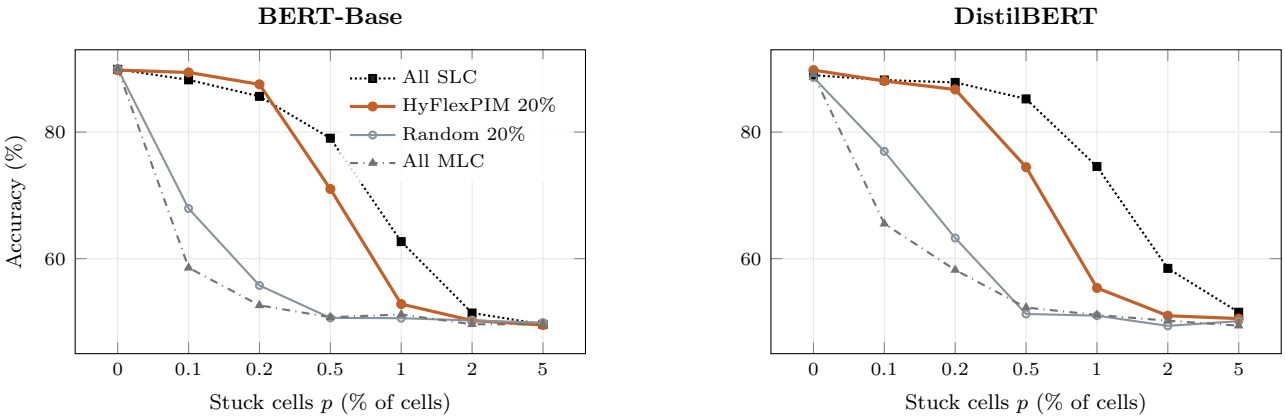


Figure 3: Accuracy against the share of stuck cells (half at $G_{\max}$, half at $G_{\min}$), no programming noise, mean of 20 chips. The horizontal axis is not to scale.

- **Stuck cells break every baseline**: all-MLC fails at 0.1% stuck cells (58.6% on BERT-Base), the HyFlexPIM hybrid drops to 71.0% at 0.5% and 52.8% at 1%, and even all-SLC falls to 62.7% at 1%.
- A stuck cell in a high-order slice shifts its weight by a large fixed amount, so faults are far more harmful than noise, and accuracy varies widely between chips depending on where the faults fall.

6 Other non-idealities

Table 4: Accuracy (%) under each other non-ideality, one at a time. The hybrid keeps 20% of the components in SLC by the HyFlexPIM gradient score.

	BERT-Base			DistilBERT		
Condition	All SLC	All MLC	HyFlexPIM	All SLC	All MLC	HyFlexPIM
No errors	89.7	90.4	90.3	88.5	89.0	88.6
Read noise $\sigma_r = 2\%$	89.0	77.3	89.8	88.3	83.8	88.1
Read noise $\sigma_r = 5\%$	80.0	53.0	76.6	84.5	57.7	80.5
$\sigma = 5\%$, middle MLC levels $\times 2$	88.2	62.5	87.8	88.4	70.3	87.1
1% stuck, scattered	61.3	50.7	52.6	74.2	51.2	55.0
1% stuck, 3×3 clusters	66.7	50.8	51.5	80.3	51.2	56.4
1% stuck, bit lines	57.0	50.4	51.0	71.2	50.8	52.2
ADC 7 bits	90.0	89.2	90.1	88.4	87.8	88.5
ADC 6 bits	89.3	83.0	89.6	88.0	81.7	88.4
ADC 5 bits	51.1	51.1	53.0	71.0	56.5	77.3
IR drop $r = 3 \times 10^{-4}$	88.9	89.8	90.5	89.4	89.1	89.6
IR drop $r = 3 \times 10^{-3}$	58.5	79.1	76.4	70.1	77.9	74.7

- Read noise and level-dependent noise behave like programming noise: all-MLC fails, and the HyFlexPIM hybrid loses 3.4–4 points against all-SLC at 5% read noise.
- A fixed 6-bit ADC is enough for all-SLC and the hybrid; all-MLC needs 7 bits; 5 bits fails BERT-Base.
- IR drop has no effect at moderate wire resistance and hurts SLC most at high resistance, because a stored SLC one sits at the highest conductance.
- Every fault pattern leaves the HyFlexPIM hybrid close to chance at 1% stuck cells.

7 Second dataset: MNLI

The same baselines on MNLI (three classes, chance 33.3%), on 2,000 matched validation pairs. On MNLI the simulator loses accuracy even on an error-free chip (up to 11 points for all-SLC on BERT-Base) because the

vector between a tier's two crossbars is re-quantised with a single 8-bit scale; only placements with the same tier sizes should therefore be compared directly.

Table 5: MNLI accuracy (%) of the baselines.

	$\sigma = 5\%$		Stuck cells (no noise)			
Placement	BERT	DistilBERT	BERT 0.1%	BERT 0.5%	DistilBERT 0.1%	DistilBERT 0.5%
Digital SVD model	83.2	80.9	83.2	83.2	80.9	80.9
All SLC	61.6	72.0	67.2	44.4	74.4	60.4
HyFlexPIM (gradient) 20%	63.3	71.3	73.0	36.9	76.0	44.5
Random 20%	35.9	44.1	42.8	33.3	51.6	32.9
All MLC	33.5	36.8	37.2	32.8	44.5	33.6

8 Summary of the baseline

- A hybrid SLC/MLC layout is necessary: all-MLC fails at moderate programming noise and at 0.1% stuck cells.
- The HyFlexPIM gradient placement works at low noise but falls behind all-SLC at $\sigma \geq 7.5\%$, by 4–16 points at 20% SLC and by up to 18 points at 5% SLC on BERT-Base.
- No baseline handles stuck cells: at 1% stuck cells the HyFlexPIM hybrid is near chance and all-SLC loses 14–27 points.

These gaps are what our contributions address; see the separate *Contribution Results* document.